■ Artificial Intelligence

C-CAT Section B — Complete Revision Sheet

Phase 2 · Topics 11 – 16

B.Tech CSE Fresher · Last-Day Rapid Revision

■ How to Use This Sheet

- Read each topic once — understand the concept, not the words.
- Exam Key Points (green boxes) = highest-priority facts for C-CAT MCQs.
- Warning boxes (yellow) = commonly confused / trick-question areas.
- Comparison tables = go-to for 'difference between X and Y' questions.
- All 6 Phase-2 topics are covered: Topics 11 → 16.

Topic 11 · AI Introduction — Definition, Agents & Environment

What is Artificial Intelligence?

AI is the field of computer science that enables machines to perform tasks that normally require human intelligence — such as reasoning, understanding language, recognising patterns, and learning from experience.

AI vs Machine Learning vs Deep Learning

These three are nested, not separate:

- AI → the broadest field: making machines 'smart'
- Machine Learning (ML) → a subset of AI; learns from data without explicit rules
- Deep Learning (DL) → a subset of ML; uses layered neural networks

> **■ AI contains ML; ML contains DL. Deep Learning is NOT broader than Machine Learning.**

AI Agent

An agent is any entity that (1) Perceives its environment through sensors, (2) Decides what to do, and (3) Acts through actuators. It does not have to be a robot — a spam filter, a chess program, and a recommendation algorithm are all agents.

PEAS Framework (Performance · Environment · Actuators · Sensors)

PEAS fully describes any AI agent. C-CAT tests this with scenario questions.

PEAS Component	Meaning	Self-Driving Car Example	Spam Filter Example
Performance	How success is measured	Safe driving, obeying rules	Correct spam classification
Environment	The world the agent operates in	Roads, traffic, weather	Email inbox
Actuators	Output tools the agent uses to ACT	Steering, brakes, accelerator	Move to spam / keep in inbox
Sensors	Input tools the agent uses to PERCEIVE	Cameras, GPS, radar	Email text, subject, sender

■ Actuators = OUTPUT (what agent DOES). Sensors = INPUT (what agent PERCEIVES). A very common trick in MCQs is to swap these two.

5 Types of AI Agents

Agent Type	How it works	Memory?	Example
Simple Reflex	Acts on current input only; follows fixed rules	No	Basic thermostat
Model-Based Reflex	Keeps internal model of world; handles partial info	Yes	Robot vacuum (remembers cleaned areas)
Goal-Based	Plans actions to reach a specific goal	Yes	GPS navigation
Utility-Based	Maximises a 'happiness score' — best possible outcome	Yes	GPS + avoids tolls + prefers highway
Learning Agent	Learns from experience and improves over time	Yes	Netflix recommendations

Order of intelligence: Simple Reflex → Model-Based → Goal-Based → Utility-Based → Learning Agent (most advanced).

Types of Environments

Property	Meaning	Example
Fully Observable	Agent can see everything	Chess (full board visible)
Partially Observable	Agent sees only part of the world	Poker (opponent cards hidden)
Deterministic	Same action → same result always	Chess move
Stochastic	Actions have uncertain outcomes	Driving in rain
Static	World doesn't change while agent thinks	Crossword puzzle
Dynamic	World changes while agent is thinking	Stock market, live traffic
Discrete	Finite number of states/actions	Chess
Continuous	Infinite range of states/actions	Self-driving car speed

Rational Agent & Turing Test

- Rational Agent: always picks the action that maximises its performance measure given what it currently knows. Does NOT mean perfect — means best possible given available info.
- Turing Test (Alan Turing, 1950): If a human cannot tell whether they are talking to a machine or another human through text, the machine is considered intelligent.

✔ AI = making machines perform human-intelligence tasks.
✔ PEAS = Performance, Environment, Actuators, Sensors.
✔ 5 agent types — Learning Agent is most advanced.
✔ Sensors = input (perceive). Actuators = output (act).
✔ Turing Test proposed by Alan Turing in 1950.
✔ Environment can be: Observable/Partial, Deterministic/Stochastic, Static/Dynamic, Discrete/Continuous.

Topic 12 · Types of AI — Purely Reactive, Limited Memory, Theory of Mind, Self-Aware

Classification by Capability

AI systems are classified based on how much they can remember, learn, and understand about themselves and others.

Type	Memory?	Understands Others?	Self-Aware?	Real-World Example
Purely Reactive (Reactive Machines)	No	No	No	IBM Deep Blue (chess), early game bots
Limited Memory	Yes — short-term only	No	No	Self-driving cars, voice assistants, ChatGPT
Theory of Mind	Yes	Yes — understands human beliefs/emotions	No	Research stage only — does not exist commercially yet
Self-Aware (Super AI / AGI)	Yes	Yes	Yes — has consciousness	Does not exist yet; theoretical

Classification by Strength (Narrow vs General vs Super)

Type	What it can do	Status	Example
Narrow AI (Weak AI)	One specific task only	Exists today	Face recognition, spam filter, AlphaGo
General AI (AGI)	Any intellectual task humans can do	Does not exist yet	Theoretical
Super AI	Surpasses human intelligence in all areas	Does not exist yet	Theoretical / sci-fi

■ **ALL AI that exists today (ChatGPT, Alexa, self-driving cars, recommendation systems) is NARROW AI. General AI and Super AI do not exist yet.**

Key Facts — Purely Reactive vs Limited Memory

- Purely Reactive: no memory at all. Reacts only to the current input. Same situation always gets the same response. Example: IBM Deep Blue.
- Limited Memory: uses recent past data + current input to make decisions. Example: self-driving car uses last few seconds of sensor data to decide actions.

- Theory of Mind AI would understand that other agents have beliefs, desires, and emotions — and use that understanding to interact. Still in research.
- Self-Aware AI would have its own consciousness and sense of self. Entirely theoretical.

✔ 4 types by capability: Reactive → Limited Memory → Theory of Mind → Self-Aware.

✔ 3 types by strength: Narrow AI (exists) → General AI → Super AI (both theoretical).

✔ All current real-world AI = Narrow AI + Limited Memory category.

✔ IBM Deep Blue = Purely Reactive. Self-driving cars = Limited Memory.

✔ Theory of Mind and Self-Aware AI do NOT exist yet.

Topic 13 · Main Domains of AI — Data Science, Computer Vision, NLP

Data Science

Data Science is the field that uses statistics, programming, and machine learning to extract useful insights and knowledge from large amounts of data.

- Key steps: Collect data → Clean data → Analyse data → Visualise → Build models → Deploy.
- Tools commonly used: Python, R, SQL, Tableau, Pandas, NumPy, Scikit-learn.
- Data Science is broader than AI — it includes statistical analysis, reporting, and visualisation that do not require AI.
- When AI/ML is applied to data, it sits within Data Science.

Computer Vision (CV)

Computer Vision is the AI domain that enables machines to interpret and understand visual information — images and videos — the way humans do.

- Core tasks: Image Classification (what is in the image?), Object Detection (where is the object?), Face Recognition, Image Segmentation.
- How it works: uses Convolutional Neural Networks (CNNs) — a type of deep learning model designed specifically for image data.
- Applications: medical imaging (detecting tumours), self-driving cars, CCTV surveillance, AR filters, quality control in manufacturing.

■ Computer Vision processes VISUAL data (images/video). NLP processes TEXT and AUDIO (language). Do not mix these up.

Natural Language Processing (NLP)

NLP is the AI domain that enables machines to understand, interpret, and generate human language — both written and spoken.

- Core tasks: Sentiment Analysis, Machine Translation, Text Summarisation, Speech Recognition, Named Entity Recognition, Question Answering.
- Key concepts: Tokenisation (splitting text into words/subwords), Stemming (reducing words to root form), Stop-word removal.
- Models used: BERT, GPT, Transformers — these are all NLP models.
- Applications: Google Translate, Siri/Alexa, chatbots, email spam filters, grammar checkers (Grammarly).

Domain	Input Type	Core Technique	Real Example
Data Science	Any data (numbers, text, images)	Statistics + ML	Sales forecasting, dashboards
Computer Vision	Images, Video	CNNs (Deep Learning)	Face unlock on phone
NLP	Text, Speech	Transformers, RNNs	Google Translate, Alexa

✔ Data Science = extract insights from data using stats + ML.

✔ Computer Vision = teach machines to 'see' (images/video).

✔ NLP = teach machines to 'understand language' (text/speech).

✔ CV uses CNNs. NLP uses Transformers/RNNs.

✔ Sentiment Analysis, Machine Translation, Speech Recognition → all NLP.

✔ Object Detection, Image Segmentation, Face Recognition → all Computer Vision.

Topic 14 · Machine Learning — Supervised, Unsupervised & Reinforcement Learning

What is Machine Learning?

Machine Learning (ML) is a subset of AI where systems learn patterns from data and improve their performance on tasks over time — without being explicitly programmed with fixed rules.

Traditional programming: Developer writes rules → Computer follows them. Machine Learning: Computer finds the rules itself by looking at examples (data).

1. Supervised Learning

The model is trained on labelled data — data where the correct answer (label) is already provided. The model learns to map inputs to outputs.

- Analogy: A student studying from a textbook that has answers at the back.
- Tasks: Classification (predicting a category) and Regression (predicting a number).
- Classification examples: spam detection (spam/not-spam), disease diagnosis (positive/negative).
- Regression examples: predicting house price, stock price forecasting.
- Common algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM (Support Vector Machine), KNN (K-Nearest Neighbours).

2. Unsupervised Learning

The model is trained on unlabelled data — no correct answers are provided. The model finds hidden patterns and structure on its own.

- Analogy: Sorting a pile of mixed objects into groups without being told the categories.
- Main tasks: Clustering (grouping similar items) and Dimensionality Reduction (simplifying data without losing too much information).
- Clustering examples: Customer segmentation, grouping news articles by topic.
- Dimensionality Reduction examples: PCA (Principal Component Analysis) — compressing many features into fewer while keeping key information.
- Common algorithms: K-Means Clustering, DBSCAN, Hierarchical Clustering, PCA.

3. Reinforcement Learning (RL)

An agent learns by interacting with an environment. It receives rewards for good actions and penalties for bad ones, and learns to maximise total reward over time.

- Analogy: Training a dog — give a treat for correct behaviour, no treat for wrong behaviour.
- Key terms: Agent (the learner), Environment (what agent interacts with), State (current situation), Action (what agent does), Reward (feedback signal), Policy (strategy).
- Examples: AlphaGo (chess/Go), game-playing AIs, robot locomotion, ad bidding systems.

Type	Data	Answer Provided?	Goal	Example
Supervised	Labelled	Yes — correct answer known	Learn input→output mapping	Email spam filter
Unsupervised	Unlabelled	No — finds own patterns	Discover hidden structure	Customer segmentation
Reinforcement	No dataset — trial & error	No — learns from rewards	Maximise cumulative reward	Game-playing AI

Key ML Terms to Know

- Overfitting: model memorises training data too well; performs poorly on new data.
- Underfitting: model is too simple; performs poorly even on training data.
- Training set / Test set / Validation set: standard split of data for building and evaluating models.
- Feature: an input variable used to make predictions (e.g. age, salary, temperature).
- Label / Target: the output variable the model is trying to predict.

■ **Supervised = LABELLED data. Unsupervised = UNLABELLED data. This is the single most tested distinction in C-CAT ML questions.**

✔ ML = systems that learn from data without explicit rules.

✔ Supervised: labelled data, learns input→output. Tasks: Classification, Regression.

✔ Unsupervised: unlabelled data, finds patterns. Tasks: Clustering, Dim. Reduction.

✔ Reinforcement: learns from rewards/penalties through trial and error.

✔ Overfitting = too well on training data, bad on new data.

✔ K-Means = Unsupervised clustering. KNN = Supervised classification.

Topic 15 · Introduction to Deep Learning — Neural Network Layers

What is Deep Learning?

Deep Learning (DL) is a subset of Machine Learning that uses Artificial Neural Networks (ANNs) with many layers to learn complex patterns from large amounts of data. The word 'deep' refers to the many layers in the network.

Structure of a Neural Network

Every neural network has three types of layers, always in this order:

Layer	Role	Analogy	Detail
Input Layer	Receives raw data	Eyes / Ears — senses the world	One neuron per feature. Does not learn; just passes data in.

Layer	Role	Analogy	Detail
Hidden Layers	Extracts patterns and features	Brain — does all the thinking	Can have 1 to hundreds of layers. More layers = deeper network = more complex patterns.
Output Layer	Produces the final prediction/result	Mouth — communicates the decision	Neurons = number of output classes (e.g. 2 for yes/no; 10 for digit 0–9).

How a Neural Network Learns

• Each connection between neurons has a weight (a number that controls importance).

• During training: data flows forward through layers — this is called Forward Propagation.

• The output is compared to the correct answer. The difference is the loss/error.

• The network adjusts weights to reduce error — this is called Backpropagation.

• This process repeats over many examples (epochs) until the model performs well.

• Activation Functions decide whether a neuron 'fires' or not. Common ones: ReLU (most common in hidden layers), Sigmoid (for binary output), Softmax (for multi-class).

Types of Neural Networks

Network Type	Full Name	Best For	Key Feature
ANN	Artificial Neural Network	General tasks	Basic feedforward structure
CNN	Convolutional Neural Network	Images, Computer Vision	Uses convolution filters to detect features in images
RNN	Recurrent Neural Network	Sequential data, NLP, Time-series	Has memory — output from previous step feeds into next
LSTM	Long Short-Term Memory	Long sequences, NLP	Improved RNN that handles long-range dependencies
GAN	Generative Adversarial Network	Generating new data	Two networks: Generator vs Discriminator

Why Deep Learning Needs Big Data

• Deep learning models have millions of parameters (weights).

• They need massive amounts of training data to learn accurately.

• This is the direct link between Phase 1 (Big Data) and Phase 2 (AI): Big Data systems store and supply the data that Deep Learning models train on.

■ CNN = images/Computer Vision. RNN/LSTM = sequences/NLP. Do not assign CNN to text tasks or RNN to image tasks.

✔ Deep Learning = subset of ML using multi-layered neural networks.

✔ 3 layers: Input (receives data) → Hidden (learns patterns) → Output (gives result).

✔ Forward Propagation = data flows forward. Backpropagation = weights are adjusted.

✔ Activation functions: ReLU (hidden layers), Sigmoid (binary), Softmax (multi-class).

✔ CNN = images. RNN/LSTM = sequences/text. GAN = generating new content.

✔ 'Deep' = many hidden layers. More layers → more complex patterns learnable.

Topic 16 · Neural Networks, Fuzzy Logic & Genetic Algorithms

Neural Networks (Recap + Deeper)

A neural network is a system inspired by the human brain. It consists of interconnected nodes (neurons) organised in layers.

- Perceptron: the simplest neural network — a single neuron. Takes inputs, multiplies by weights, adds a bias, applies activation function.
- Multi-Layer Perceptron (MLP): multiple perceptrons in layers — the basis of most ANNs.
- Epoch: one full pass through the entire training dataset.
- Batch: a small subset of training data used in one weight update step.
- Learning Rate: controls how large each weight update step is. Too high → unstable. Too low → very slow learning.
- Dropout: a regularisation technique — randomly 'turns off' some neurons during training to prevent overfitting.

Fuzzy Logic

Traditional (Boolean) logic: everything is either TRUE (1) or FALSE (0). Fuzzy Logic allows degrees of truth — values between 0 and 1.

- Invented by Lotfi Zadeh in 1965.
- Analogy: Traditional logic says temperature is either 'hot' or 'not hot'. Fuzzy logic says temperature can be '70% hot' or '30% hot'.
- Membership Function: defines how much an input belongs to a fuzzy category.
- Applications: washing machines (how dirty is 'dirty'?), air conditioners, anti-lock braking systems (ABS), medical diagnosis systems.
- Why it matters: real-world data is rarely perfectly black-and-white. Fuzzy logic handles vagueness and uncertainty better than crisp logic.

Genetic Algorithms (GA)

Genetic Algorithms are optimisation techniques inspired by Darwin's theory of natural evolution — survival of the fittest applied to finding the best solution to a problem.

- Key idea: start with a population of candidate solutions; evolve them over generations toward better solutions using biological operators.
- Chromosome: one candidate solution (like a 'string' of potential answers).
- Fitness Function: evaluates how good a solution is. Higher fitness = better solution.
- Selection: better-performing solutions are chosen to 'reproduce'.
- Crossover (Recombination): two parent solutions are combined to create offspring.
- Mutation: small random changes are introduced to maintain diversity.
- Applications: route optimisation (Travelling Salesman Problem), scheduling, game AI strategy, engineering design, hyperparameter tuning in ML.

Concept	Core Idea	Inspired By	Best Used For
Neural Networks	Layered interconnected nodes that learn from data	Human brain neurons	Pattern recognition, prediction, classification
Fuzzy Logic	Degrees of truth (0 to 1) instead of strict True/False	Human vague reasoning	Control systems, imprecise decision-making
Genetic Algorithm	Evolve solutions through selection, crossover, mutation	Biological evolution	Optimisation problems with many possible solutions

Genetic Algorithm — Step-by-Step Process

Step	What happens
1. Initialisation	Create a random population of candidate solutions
2. Fitness Evaluation	Score each solution using the fitness function
3. Selection	Pick the best solutions to be parents
4. Crossover	Combine pairs of parents to create offspring
5. Mutation	Randomly change parts of some offspring
6. New Generation	Replace old population with new offspring
7. Repeat	Repeat steps 2–6 until good solution is found or limit reached

■ **Fuzzy Logic was invented by Lotfi Zadeh (1965). Genetic Algorithms were popularised by John Holland (1970s). C-CAT may ask about founders/years.**

- ✔ Perceptron = single neuron. MLP = multiple layers of perceptrons.
- ✔ Epoch = one full pass through training data. Learning Rate = step size of weight updates.
- ✔ Fuzzy Logic: truth has degrees (0 to 1). Invented by Lotfi Zadeh, 1965.
- ✔ Membership Function = defines how much input belongs to a fuzzy category.
- ✔ Genetic Algorithm: Selection → Crossover → Mutation (3 main operators).
- ✔ Fitness Function = measure of how good a GA solution is.
- ✔ GA is used for OPTIMISATION problems (finding best solution among many).

■ Master Quick-Reference — All Topics at a Glance

Topic	1-Line Summary	Must-Remember Fact
11 – AI Intro	AI = machines doing human-intelligence tasks	PEAS: Performance·Environment·Actuators·Sensors
11 – Agents	Agent = perceives + decides + acts	5 types: Reactive→Model→Goal→Utility→Learning
12 – Types of AI	Classified by capability and by strength	All current AI = Narrow AI + Limited Memory
13 – Data Science	Extract insights using stats + ML	Broader than AI; includes visualisation too
13 – Computer Vision	AI that understands images/video	Uses CNNs; Object Detection, Face Recognition
13 – NLP	AI that understands human language	Sentiment Analysis, Translation, Speech Recog.
14 – Supervised ML	Learns from labelled data	Classification (category) + Regression (number)
14 – Unsupervised ML	Finds patterns in unlabelled data	Clustering (K-Means) + Dimensionality Reduction (PCA)
14 – Reinforcement	Learns via rewards/penalties	Agent · Environment · State · Action · Reward
15 – Deep Learning	Multi-layered neural networks	Input → Hidden → Output layers

Topic	1-Line Summary	Must-Remember Fact
15 – CNN	For images	Convolutional filters detect visual features
15 – RNN/LSTM	For sequences	Has memory; used in NLP and time-series
16 – Fuzzy Logic	Degrees of truth, not just 0 or 1	Lotfi Zadeh, 1965. Membership Function.
16 – Genetic Algo.	Evolves solutions like natural selection	Selection → Crossover → Mutation → New Generation

■ Top 10 Commonly Confused Points — C-CAT Traps

Trap 1: Sensors vs Actuators	Sensors = INPUT (perceive). Actuators = OUTPUT (act). MCQs often swap these — always ask: 'Is the agent receiving info or doing something?'
Trap 2: AI ⊃ ML ⊃ DL	AI is the largest category. ML is inside AI. DL is inside ML. Never say DL is broader than ML.
Trap 3: Supervised vs Unsupervised	Supervised = LABELLED data (you provide answers). Unsupervised = UNLABELLED data (model finds its own answers). The label is the deciding factor.
Trap 4: CNN vs RNN	CNN = images/Computer Vision. RNN/LSTM = sequences/NLP/time-series. Do not mix.
Trap 5: Narrow AI vs General AI	All current AI (ChatGPT, Alexa, self-driving) = Narrow AI. General AI does not exist yet.
Trap 6: Purely Reactive vs Limited Memory	Purely Reactive = NO memory at all (e.g. Deep Blue). Limited Memory = uses recent past data (e.g. self-driving car).
Trap 7: Classification vs Regression	Classification = output is a CATEGORY (spam/not-spam, disease positive/negative). Regression = output is a NUMBER (house price, temperature).
Trap 8: Overfitting vs Underfitting	Overfitting = too good on training data, bad on new data (memorised, not learned). Underfitting = too simple, bad even on training data.
Trap 9: Fuzzy Logic: invented by	Lotfi Zadeh, 1965. Not to be confused with Genetic Algorithms (John Holland, 1970s).
Trap 10: GA Operators	Genetic Algorithm has 3 operators: Selection, Crossover, Mutation. Missing even one in an MCQ makes the answer wrong.

Best of luck tomorrow! You have put in the work — trust your preparation. ■ For C-CAT, recognise the concept and eliminate wrong options. You've got this.